



CS 402: Compiler Technology

Course Code	CS 402
Course Name	Compiler Technology
L-T-P-C	0-0-3-2
Prerequisites	None Algorithms and Data Structures; Formal Languages and Automata Theory; Paradigms of Programming
Intended for	B.Tech.
Distribution	Compulsory for CSE; CS elective for EE and ME
One-time approval	5th Senate (Oct 2013)

Course Contents

- **Moduel I** Introduction to compilers, Lexical Analysis, Syntax Analysis, Parsing Top down, Bottom up and advanced, Syntax directed translation, Intermediate Code Generation type checking and control flow.
- **Moduel II** Run time environments Stack allocation, Heap management, Garbage Collection, Code generation and Optimization machine independent, machine dependent, parallelism, data flow analysis and locality.
- **Moduel III** LLVM - Introduction, Illustration with examples, Mini-project.
- **Moduel IV** Compiling Functional Languages Review of Lambda Calculus, Translating functional programs into lambda calculus, Program representation and Graph reduction of lambda expressions, Super-combinators and lambda lifting, Advanced Graph Reduction the G-Machine and Optimizations.
- **Moduel V** (Optional time permitting) Compiling with Continuations Review of Continuations and CPS, Conversion to CPS, Optimization of CPS, Closure Conversion and Machine code generation.

Text Books

1. A. Appel, **Modern Compiler Implementation in C (Java, ML)**, Cambridge Univ. Press.
2. Simon Peyton-Jones, **Implementation of Functional Languages**, Prentice-Hall.
3. A. Appel, **Compiling with Continuations**, Cambridge University Press.

References

1. Aho, Lam, Sethi and Ullman, **Compilers Principles, Techniques, Tools**, 2nd Edition, Pearson/Addison-Wesley.